

Answers to Exercise 1:

For two consecutive experiments, the total number of possible outcomes is calculated using the addition principle:

$$\text{Total outcomes} = \sum_{i=1}^m n_i$$

where m is the number of outcomes for the first experiment and n_i is the number of outcomes for the second experiment corresponding to each outcome of the first.

Answers to Exercise 2:

To find the possible initials consisting of 2 letters:

$$\text{Total initials} = 26 \times 26.$$

To guarantee at least two students share the same initials, at least $26^2 + 1$ students are needed in a class.

Answers to Exercise 3:

For the 10 volumes of an encyclopedia arranged randomly:

- Total arrangements without restrictions:
 $10!$
- If volumes 7, 6, and 5 must be placed next to each other in that specific order, treat them as a single unit:
 $8!$

Answers to Exercise 4:

1. Total 3-letter words with no restrictions:

$$26^3.$$

2. Total with at least 2 consonants:

$$\text{Total} = (3 \text{ consonants}) \text{ or } (2 \text{ consonants, 1 vowel}).$$

$$21^3 + 3 \cdot 21^2 \cdot 5$$

3. Exactly 2 consonants and 1 vowel:

$$21^2 \cdot 5 \cdot 3,$$

select 2 consonants, 1 vowel, and arrange them (ccv or cvc or vcc).

4. At most 1 vowel:

$$\text{Total} = \text{no vowel or } (2 \text{ consonants, 1 vowel}) = (3 \text{ consonants}) \text{ or } (2 \text{ consonants, 1 vowel}).$$

$$21^3 + 3 \cdot 21^2 \cdot 5$$

Same as question 2.

Answers to Exercise 5:

1. Total ways to visit 7 family members on the same day:

$$7!.$$

2. If 4 are visited on Eid and 3 the next day:

$$(7)_4 \cdot 3! = 7!.$$

Answers to Exercise 6:

1. Total football teams from 22 people:

$$\binom{22}{11}.$$

2. Teams formed with 3 goalkeepers including at least 1:

$$\text{Choose 1 goalkeeper and 10 from 19 others} \Rightarrow 3 \cdot \binom{19}{10}.$$

Answers to Exercise 7:

A company with 2 managers, 3 administrators, and 11 workers:

1. Total ways without restrictions:

$$\binom{16}{4}.$$

2. 1 manager, 2 administrators, and 1 worker:

$$\binom{2}{1} \cdot \binom{3}{2} \cdot \binom{11}{1}.$$

3. Workers and administrators cannot be together:

- Solution 1: Choose them from ($\text{admins} \cup \text{managers}$) or ($\text{workers} \cup \text{managers}$):

$$\binom{3+2}{4} + \binom{11+2}{4} = \binom{5}{4} + \binom{13}{4}$$

- Solution 2: Consider all the cases:

$$\left. \begin{array}{l} 4W : \binom{11}{4} \\ 3W, 1M : \binom{11}{3} \binom{2}{1} \\ 2W, 2M : \binom{11}{2} \binom{2}{2} \\ 3A, 1M : \binom{3}{3} \binom{2}{1} \\ 2A, 2M : \binom{3}{2} \binom{2}{2} \end{array} \right\} \xrightarrow{A.P.} \text{Total} = \binom{11}{4} + \binom{11}{3} \binom{2}{1} + \binom{11}{2} \binom{2}{2} + \binom{3}{3} \binom{2}{1} + \binom{3}{2} \binom{2}{2}$$

- Solution 3: We consider all the total number - the number where admins and workers meet ($3W, 1A$), ($2W, 2A$), ($2W, 1A, 1M$), ($1W, 3A$), ($1W, 2A, M$), ($1W, 1A, 2M$):

$$\binom{16}{4} - \left(\binom{2}{2} \binom{3}{1} \binom{11}{1} + \binom{2}{1} \binom{3}{2} \binom{11}{1} + \binom{3}{3} \binom{11}{1} + \binom{2}{1} \binom{3}{1} \binom{11}{2} + \binom{3}{2} \binom{11}{2} + \binom{3}{1} \binom{11}{3} \right).$$

Answers to Exercise 8:

To divide a class of 100 students into groups of sizes 15, 18, 20, 21, and 26, we can use the binomial coefficient approach:

$$\text{Total choices} = \binom{100}{15} \times \binom{85}{18} \times \binom{67}{20} \times \binom{47}{21}$$

The last group will consist of the remaining 26 students, which is represented as $\binom{26}{26} = 1$.

Thus, the total number of choices for this division is given by:

$$\text{Total choices} = \binom{100}{15} \times \binom{85}{18} \times \binom{67}{20} \times \binom{47}{21} = \frac{100!}{15! \cdot 18! \cdot 20! \cdot 21! \cdot 26!}$$

Answers to Exercise 9:

In a committee of 10 people, a chairperson, a secretary and a treasurer should be selected

1. Knowing that accumulation is not allowed, in how many ways can these offices be allocated if

- i. There are no restrictions?

Answer: $(10)_3 = \binom{10}{3} 3! = 720$

- ii. A and B cannot be together?

Answer 1: $\bar{A}\bar{B} \vee \bar{A}B \vee A\bar{B}$

$$\binom{8}{3} 3! + \binom{1}{1} \binom{8}{2} 3! + \binom{1}{1} \binom{8}{2} 3!$$

Answer 2: $\bar{A} \vee A\bar{B}$

$$\binom{9}{3} 3! + \binom{1}{1} \binom{8}{2} 3!$$

Answer 3: $Total - AB$

$$\binom{10}{3} 3! - \binom{2}{2} \binom{8}{1} 3!$$

- iii. C and D are together or not at all?

Answer: $CD + \bar{C}\bar{D}$:

$$\binom{2}{2} \binom{8}{1} 3! + \binom{8}{3} 3! = 384$$

- iv. F must have a task?

Answer:

$$\binom{2}{2} \binom{8}{1} 3! + \binom{8}{3} 3! = 216$$

- v. F only accepts the task of chairperson?

Answer: F president or not all

$$\binom{9}{2} 2! + \binom{9}{3} 3!$$

2. Knowing that accumulation is allowed, in how many ways can these offices be allocated if there are no restrictions?

Answer:

$$10^3$$

Answers to Exercise 10:

- If the student has to answer 5 out of 8 questions:

$$\binom{8}{5} = 56$$

- If the student has to answer at least 3 out of the first 5 questions: So he exactly 3 or 4 or 5 out of the first 5 questions and 2 from the remaining 3:

$$\binom{5}{3} \times \binom{3}{2} + \binom{5}{4} \times \binom{3}{1} + \binom{5}{5} = 46$$

Answers to Exercise 11:

- For the word "ALGER":

$$5! = 120$$

- For the word "ANNABA":

$$\frac{6!}{3! \times 2!} = 60$$

- For the word "TIZIOUZOU":

$$\frac{9!}{(2!)^4}$$

- For the word "MISSISSIPPI":

$$\frac{11!}{4! \times 4! \times (2!)} = 34,650$$

Answers to Exercise 12:

In how many ways can 8 people be seated?

- In a row,

1. **If there are no restrictions:** The total arrangements of 8 people is given by:

$$8!$$

2. **Two persons A and B stay together:** Treat A and B as a single unit. This creates 7 units (AB and 6 others):

$$7! \times 2!$$

3. **There are 4 adults and 4 children, each adult has only children as neighbors and vice versa:** Arrange adults and children alternatively:

Arrangement: *ACACACAC* or *CACACACA*.

Total arrangements:

$$4! \times 4! \times 2.$$

4. **The number of adults is 6, they must stay together:** Treat the 6 adults as a single unit. This creates 3 units (ADULTS and 2 children):

$$3! \times 6!$$

5. **There are 4 children, each accompanied by his mother who stays next to him:** Treat each mother-child pair as a single unit. This creates 4 units (2! permutations of each pair):

$$4! \cdot (2!)^4.$$

- Around a table

1. **If there are no restrictions:**

$$7!$$

3. **There are 4 adults and 4 children, each adult has only children as neighbors and vice versa:**

$$3! \times 4!$$

Answers to Exercise 13:

Group the athletes by nationality:

- Algerian block (4 athletes)
- Libyan block (3 athletes)
- Tunisian block (2 athletes)

$$3! \times 4! \times 3! \times 2!$$

Exercise 1. p balls numbered from 1 to p are placed in n indistinguishable boxes ($p \geq n$). Let D_n^p be the number of possible arrangements such that no box is empty.

1. Show that $D_n^p = nD_n^{p-1} + D_{n-1}^{p-1}$ $p \geq n + 1$.
2. Calculate a/ D_1^p and D_n^n b/ D_2^p and D_3^p c/ D_{p-1}^p , D_{p-2}^p and D_{p-3}^p .

Solution:

1. We show that $D_n^p = nD_n^{p-1} + D_{n-1}^{p-1}$, $p \geq n + 1$, using a combinatorial argument. Let's consider the $p - th$ ball and see how it can contribute to the distribution of the balls.

Case 1: The $p - th$ ball is in a box by itself.

If the $p - th$ ball is alone in a box, then the remaining $p - 1$ balls must be distributed into $n - 1$ boxes (since we have already used one box for the $p - th$ ball). The number of ways to do this is D_{n-1}^{p-1} .

Case 2: The $p - th$ ball is not by itself.

First, we distribute the $p - 1$ balls into n boxes in D_n^{p-1} ways, and there are n ways to choose a box for the $p - th$ ball, by (MP), the total numbers of ways is nD_n^{p-1} .

By (AP), the number of ways to distribute p balls into n boxes such that no box is empty is,

$$D_n^p = nD_n^{p-1} + D_{n-1}^{p-1}.$$

2. a/ $D_1^p = 1$, which represents the number of ways to distribute p balls into 1 box. There is only one way to do this: by placing all the balls into the single box.
 b/ $D_n^n = 1$, which represents the number of ways to distribute n balls into n boxes such that each box contains exactly one ball. There is only one way to do this: each ball must go into a box.
 c/ D_2^p : 1st approach: To find D_2^p , we can proceed with a combinatorial argument. We want to distribute a set of p balls, say $\{1, 2, \dots, p\}$, into exactly 2 non-empty boxes. Let's call these two boxes A and B . We need to count the number of ways to form such distribution.
 For each distribution, one box must contain k balls, and the other box must contain the remaining $p - k$ balls, where $1 \leq k \leq p - 1$.

Once we decide on k , there are $\binom{p}{k}$ ways to choose which k balls will be in box A . The remaining $p - k$ balls will automatically go into box B .

Since the two boxes are unordered (i.e., A and B are the same distribution as B and A), each choice has been counted twice. Therefore, we need to divide by 2 to avoid double-counting.

Putting it all together, we sum over all possible sizes k for one of the boxes:

$$\begin{aligned} D_2^p &= \frac{1}{2} \sum_{k=1}^{p-1} \binom{p}{k} \\ &= \frac{2^p - 2}{2} = 2^{p-1} - 1. \end{aligned}$$

2nd approach: To determine D_2^p , that counts the ways to distribute a set of p balls into exactly 2 non-empty, unordered boxes, we know the fact that the total number of ways to assign p balls to two boxes is 2^p . For each of the p balls, there are 2 choices: it can either go into box A or box B . Thus, there are 2^p ways to assign p balls to two labeled boxes (where boxes A and B are distinct).

In our total count 2^p , we include cases where all balls end up in box A (leaving B empty) or all balls end up in box B (leaving A empty). There are exactly 2 such cases. Therefore, the number of ways to assign the p balls to two boxes such that neither box is empty is:

$$2^p - 2.$$

Since we only care about distribution of p balls into two unordered boxes, we have counted each valid distribution twice: once as (A, B) and once as (B, A) . To correct for this, we divide by 2.

So, the number of ways to distribute p balls into exactly 2 non-empty, unordered boxes is:

$$D_2^p = \frac{2^p - 2}{2} = 2^{p-1} - 1.$$

d/ To determine D_3^p , that counts the number of ways to distribute a set of p balls into exactly 3 non-empty, unordered boxes.

For each of the p balls, there are 3 choices: it can go into box A , B , or C . Therefore, there are 3^p ways to assign p balls to three labeled boxes (A, B, C) .

We need exactly 3 non-empty boxes.

Suppose one box (say A) is empty. Then, each of the p balls has only 2 choices: it can go into either B or C . So, there are 2^p ways for all balls to be assigned to either B or C with A empty. The same count 2^p applies if B or C is empty.

Since there are 3 ways to choose which of A , B , or C is empty, the total number of ways where at least one box is empty is:

$$3 \cdot 2^p.$$

If two boxes are empty (say A and B), then all balls must go into the remaining box C . There is only 1 way to do this (all balls in C), so there are 3 ways total (one for each pair of empty boxes: (A, B) , (A, C) , (B, C)).

Thus, we add back 3 to account for the cases where two boxes are empty.

the number of ways to assign balls to the three boxes such that no box is empty is:

$$3^p - 3 \cdot 2^p + 3.$$

The calculation above counts each distribution as ordered (i.e., $\{A, B, C\}$ is treated differently from $\{B, A, C\}$), while we only care about unordered distributions. Since there are $3! = 6$ ways to order 3 boxes, we divide by 6 to avoid overcounting.

Thus, the number of ways to distribution p balls into exactly 3 non-empty, unordered boxes is:

$$D_3^p = \frac{1}{6} (3^p - 3 \cdot 2^p + 3).$$

e/ To find D_{p-1}^p , the number of ways to distribute a set of p balls into exactly $p - 1$ non-empty, unordered boxes, we can proceed with a combinatorial argument.

If we want to distribute p balls into $p - 1$ boxes, then exactly one of these boxes must contain 2 balls, while the remaining $p - 2$ boxes must contain 1 ball each. This is the only possible configuration for this type of distribution.

To create a distribute into $p - 1$ boxes, we first choose 2 balls out of the p balls to form a box with 2 balls. The number of ways to select 2 balls from p is given by $\binom{p}{2}$.

After choosing 2 balls to form a box of size 2, each of the remaining $p - 2$ balls must be placed in its own box. There is only 1 way to do this, as each ball is in a box by itself.

$$D_{p-1}^p = \binom{p}{2} = \frac{p(p-1)}{2}.$$

f/ To find D_{p-2}^p , the number of ways to distribute a set of p balls into exactly $p - 2$ non-empty, unordered boxes, we can proceed with a combinatorial argument. There are

two distinguishable cases, either there is a box that contains 3 balls or there are two boxes that contain 2 balls each.

Case 1: One box contains 3 balls

This can be done in $\binom{p}{3}$ ways (see the previous question). The remaining $p - 3$ balls each will be in a box by itself. There's only one way to do this.

Case 2: Two boxes with 2 balls each,

choose 2 balls for the first box: in $\binom{p}{2}$ ways, then choose 2 balls from the remaining $p - 2$ for the second box in $\binom{p-2}{2}$ ways. The remaining $p - 4$ balls each is in one box, there is only one way.

So, the number of distributions in this case is $\binom{p}{2} \binom{p-2}{2}$.

When we choose 2 elements for the first box and 2 for the second, we're essentially labeling the boxes. But in reality, the order of the boxes doesn't matter. For example, choosing {1, 2} and {3, 4} is the same as choosing {3, 4} and {1, 2}. So, we've counted each distribution twice. To correct this, we divide the number of distributions by 2. Thus,

$$D_{p-2}^p = \binom{p}{3} + \binom{p}{2} \binom{p-2}{2} \frac{1}{2}$$

g/ D_{p-3}^p See if you can solve this one yourself.

1. $\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}$ The left-hand side counts the ways to select a group of k people chosen from a set of n people and then to select a subset of r leaders within the group of k people. Equivalently, as counted on the right side, we could first select the subset of r leaders from the set of n people and then select the remaining $k-r$ members of the group from the remaining $n-r$ people.
2. Using the previous identity $\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}$, the LHS becomes:

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} = \binom{n}{r} \sum_{k=r}^n \binom{n-r}{k-r}.$$

Now, let $m = k - r$. When $k = r$, $m = 0$, and when $k = n$, $m = n - r$. Changing the summation index gives:

$$\sum_{k=r}^n \binom{n-r}{k-r} = \sum_{m=0}^{n-r} \binom{n-r}{m}.$$

Using the identity $\sum_{m=0}^n \binom{n}{m} = 2^n$, we have:

$$\sum_{m=0}^{n-r} \binom{n-r}{m} = 2^{n-r}.$$

Thus, the LHS becomes:

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} = \binom{n}{r} \cdot 2^{n-r}.$$

3. From a group of n people, there are $\binom{n}{k}$ ways to form a k -member committee, and k ways to select a leader from the k members in the committee. Thus, the number of ways to form a k -member committee including the selection of a leader is, by the multiplication principle, $k \binom{n}{k}$. As k could be $1, 2, \dots, n$, by the addition principle, the number of ways to do so is given by

$$\sum_{k=0}^n k \binom{n}{k}.$$

Let us count the same problem via a different approach as follows. First, we select a leader from the group, and then choose $k - 1$ members, where $k = 1, 2, \dots, n$, from the group to form a k -member committee. There are n choices for the first step and

$$\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{n-1},$$

ways for the second step. Thus, by the multiplication principle, the required number is

$$n \left[\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{n-1} \right] = n2^{n-1}.$$

Since both count the same number, we have

$$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}.$$

4.

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} (-1)^{n-k} = 0,$$

Using the identity

$$\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r},$$

the summation becomes:

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} (-1)^{n-k} = \binom{n}{r} \sum_{k=r}^n \binom{n-r}{k-r} (-1)^{n-k}.$$

Let $j = k - r$. Then when $k = r$, $j = 0$, and when $k = n$, $j = n - r$. Changing the summation index gives:

$$\sum_{k=r}^n \binom{n-r}{k-r} (-1)^{n-k} = \sum_{j=0}^{n-r} \binom{n-r}{j} (-1)^{n-(j+r)} = (-1)^{n-r} \sum_{j=0}^{n-r} \binom{n-r}{j} (-1)^j.$$

The alternating sum of binomial coefficients satisfies the identity:

$$\sum_{j=0}^m \binom{m}{j} (-1)^j = 0, \quad \text{for } m > 0.$$

Here, $m = n - r > 0$ (since $r \leq n$). Thus:

$$\sum_{j=0}^{n-r} \binom{n-r}{j} (-1)^j = 0.$$

Since the alternating sum vanishes, we have:

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} (-1)^{n-k} = \binom{n}{r} \cdot 0 = 0.$$

Thus, the identity

$$\sum_{k=r}^n \binom{n}{k} \binom{k}{r} (-1)^{n-k} = 0.$$

holds.

5. Suppose there is n students and we want to choose k from them, we can do that in $\binom{n}{k}$ ways. Another way to do that is to consider two students, either we choose both of them, now we need to choose $k-2$ students from $n-2$ (because we already chose two) in $\binom{n-2}{k-2}$ ways, or we choose one of them in $\binom{2}{1}$ ways and we complete by choosing $\binom{n-2}{k-1}$, the third case is we don't choose both of them, so we have $\binom{n-2}{k}$ ways to do that. In total we have $\binom{n-2}{k-2} + 2\binom{n-2}{k-1} + \binom{n-2}{k}$, both sides count the same number, so we have

$$\binom{n}{k} = \binom{n-2}{k-2} + 2\binom{n-2}{k-1} + \binom{n-2}{k}.$$

Combinatorial Proof of the Hockey Stick Identity

$$\sum_{j=k}^n \binom{j}{k} = \binom{n+1}{k+1}$$

Combinatorial proof

Let's consider a group of $n + 1$ people. We want to choose a committee of $k + 1$ people from this group, and we want to count the number of ways to do this in two different ways.

Method 1: Directly choose $k + 1$ people from the $n + 1$ people. This can be done in $\binom{n+1}{k+1}$ ways.

Method 2: We can choose the oldest person on the committee first. Once we've chosen the oldest person, we need to choose k more people from the remaining n people. Note that we start with the $(k + 1)^{th}$ person. This ensures we have at least k people younger to choose from. If the oldest person is the $(k + 1)^{th}$ person, we choose k people from k people, which can be done in $\binom{k}{k}$ ways.

If the oldest person is the $(k + 2)^{th}$ person, we choose k people from $k + 1$ person, which can be done in $\binom{k+1}{k}$ ways. ... If the oldest person is the $(n + 1)^{th}$ person, we choose k people from n people, which can be done in $\binom{n}{k}$ ways. So, the total number of ways to choose the committee using this method is

$$\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k} = \sum_{j=k}^n \binom{j}{k}$$

Since both methods count the same thing, the two expressions must be equal:

$$\sum_{j=k}^n \binom{j}{k} = \binom{n+1}{k+1}$$